

SMART APS

Advanced Planning and Scheduling System

Plain Language Guide to Constraints and Solutions

Document Title	Smart APS — Plain Language Constraints Guide
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This document explains, in plain language, every constraint built into the Smart APS system — what the problem is, how the system solves it, and what happens when the primary solution is not sufficient. No technical background is required to understand this document.

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1. INTRODUCTION AND PURPOSE

Smart APS (Advanced Planning and Scheduling System) is a software tool developed in-house by the Production Planning Department to automate the daily production plan for the Injection Moulding Division. It runs each morning, reads the current inventory levels and monthly targets from existing Excel files, and produces a complete, ready-to-execute daily plan within sixty seconds.

Every good planning system operates within a set of rules, called constraints. These constraints exist because the real factory floor has real limitations: machines can only run for so many hours, only one changeover can happen at a time, and some parts are more urgent than others. Without these rules, a plan would be mathematically possible on paper but impossible to execute in practice.

This document explains each of the nine constraints built into Smart APS in plain language. For each constraint, the following structure is used throughout:

THE PROBLEM	What goes wrong without this rule.
OUR SOLUTION	What Smart APS does to prevent the problem.
IF IT STILL FAILS	The backup plan when the primary solution is not sufficient.

2. CONSTRAINT 1: THE 22-HOUR SHIFT RULE

The Problem

Each machine operates for a maximum of 22 hours in a single shift. Without a strict rule, a planner could accidentally assign more work to a machine than it can physically complete. The extra parts would never be made, and nobody would know until the shift was already underway. This leads to missed production targets and emergency replanning mid-shift.

Our Solution

Before assigning any part to a machine, Smart APS checks exactly how many hours have already been allocated to that machine. It calculates the run time required for the part and the time needed for the mould changeover, adds them together, and only proceeds if the total fits within the remaining hours. The machine can never be over-committed.

The logic the system follows is shown in Figure 1 below.

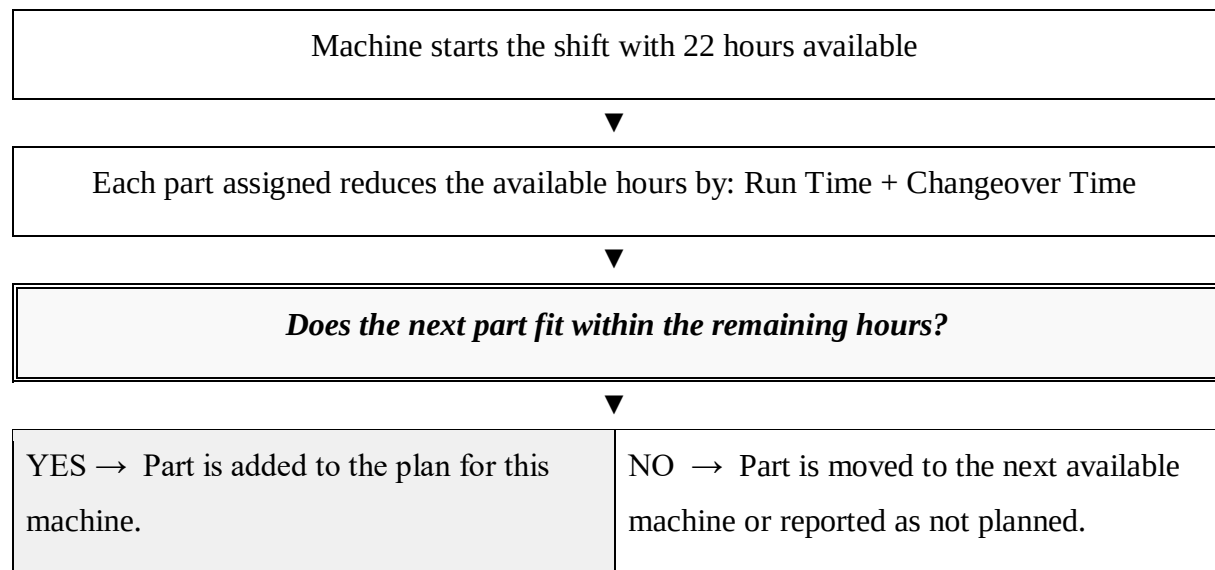


Figure 1: Decision flow for the 22-hour shift rule.

If It Still Fails

If a part cannot fit on any compatible machine that day, it is placed on a report called VT_Not_Planned. This report shows the exact reason the part could not be scheduled and the precise number of free hours remaining on each compatible machine. The planner can then decide whether to adjust quantities of lower-priority parts to create space, or defer the part to the following day.

3. CONSTRAINT 2: MINIMUM RUN BLOCK OF FOUR HOURS

The Problem

Setting up a mould on an injection moulding machine (called a changeover) takes approximately 40 minutes. If a part is only planned to run for one or two hours after this setup, the changeover time represents a very large proportion of the total time used, and very few parts are produced. This makes poor use of the machine and the tool-change team's time.

Our Solution

Smart APS will only assign a part to a machine if, after subtracting the changeover time, at least four hours remain for actual production. If only three hours are available after the changeover, the system does not assign that part to that machine. It moves on to find another machine that has more capacity. Figure 2 illustrates this check.

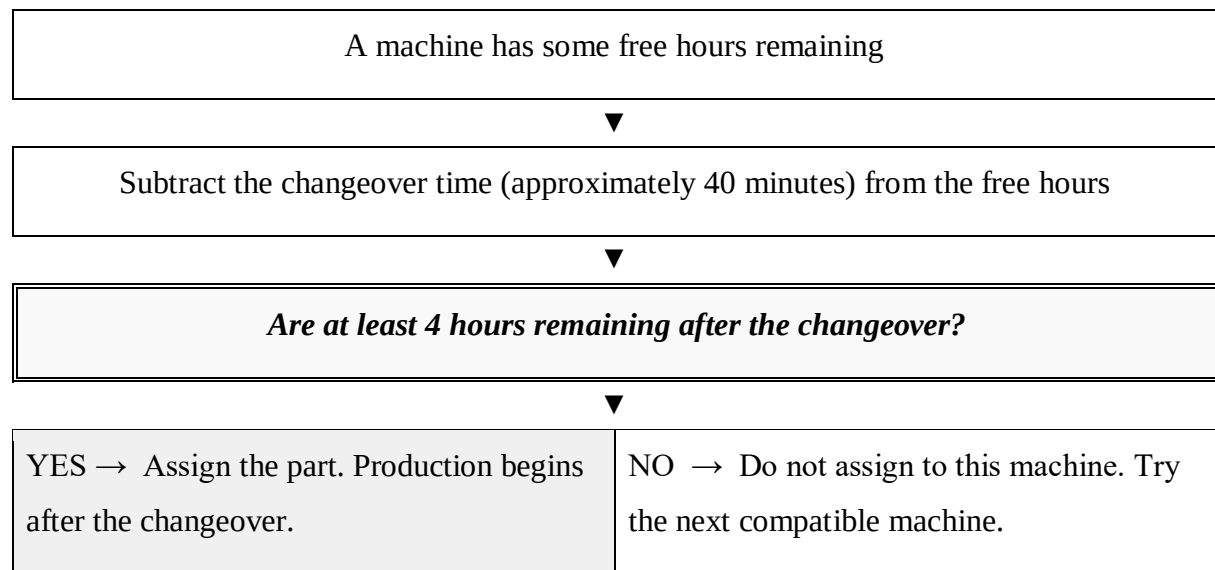


Figure 2: Minimum run block check.

If It Still Fails

If no machine has four or more effective hours available for a particular part, that part is added to the VT_Not_Planned report. The planner can review it and, if necessary, reduce the planned quantity of another part on a suitable machine to free up enough time.

4. CONSTRAINT 3: ONE TOOL CHANGER FOR ALL 30 MACHINES

The Problem

There is a single shared tool-change team that performs all mould changeovers across the entire VT machine floor. They can only work on one machine at a time. If the plan requires several machines to undergo a changeover at the same moment, the tool changers can only attend to one of them. The others must wait, sitting completely idle during that time. This idle time is wasted production capacity.

Our Solution

After the main plan is built, Smart APS collects every changeover event across all machines and sequences them so that no two changeovers overlap in time. This is done using a scheduling algorithm that finds the optimal order. Furthermore, when a machine must wait because the tool changers are busy elsewhere, the system automatically extends the run of the current part on that machine to fill the waiting time. The machine continues producing instead of standing idle.

Figure 3 shows how this works.

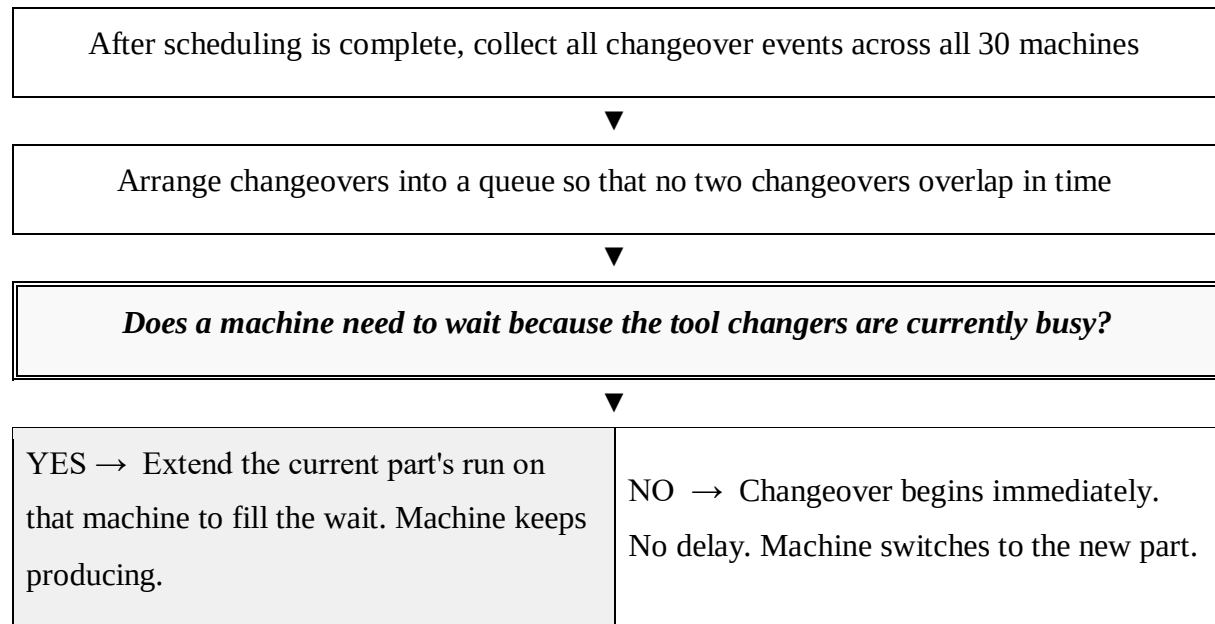


Figure 3: Tool-changer sequencing and machine wait resolution.

If It Still Fails

The scheduling algorithm has a ten-second time limit. If it cannot find the perfect sequence within ten seconds (which can happen when there are many simultaneous changeovers), it automatically switches to a faster method that still ensures no two changeovers overlap, though it may not be the mathematically perfect sequence. In either case, the plan is always completed on time and the no-overlap rule is never broken.

5. CONSTRAINT 4: CONTINUE THE SAME PART TO AVOID CHANGEOVER

The Problem

If a machine ran Part A at the end of yesterday's shift, and today's plan assigns a different part first, a 40-minute mould changeover is required before any production can begin. Across 30 machines, if this happens on each one every day, hundreds of minutes of production time are lost to unnecessary changeovers. This is avoidable if the same part is simply continued from where it left off.

Our Solution

At the end of each day, Smart APS saves the last part run on each machine to a file called machine_state.json. The following morning, when ranking which machine to assign a part to, the system gives a strong preference to the machine that was already running that part. If the same machine is chosen, the mould is already loaded and production begins immediately at the start of the shift with no changeover. Table 1 below summarises the two scenarios.

	Scenario A: Same Part as Yesterday	Scenario B: Different Part from Yesterday
Mould status	Already loaded and ready	Must be removed and replaced
Changeover needed	No	Yes, approximately 40 minutes
Production start	Immediately at shift start (00:00)	After the changeover is complete
Time saving	40 minutes saved per machine	40 minutes consumed per machine

Table 1: Comparison of same-part continuation versus changeover.

If It Still Fails

If the machine that previously ran a part is no longer available (for example, it is fully occupied with other parts), Smart APS assigns the next best available machine and includes the 40-

minute changeover honestly in the plan. This changeover is then automatically sequenced by the tool-changer scheduler described in Constraint 3.

6. CONSTRAINT 5: SPECIALISED MACHINES MUST BE PROTECTED

The Problem

Some machines on the factory floor can only run two or three specific parts. These are called specialised machines. If a general-purpose part is assigned to a specialised machine and uses up its available hours, the parts that only that machine can produce have nowhere to go for the entire day. They cannot be made, even though the right machine exists. This creates avoidable stock-outs for those specific parts.

Our Solution

Smart APS reads a data sheet called VT_Machine_Part_Count, which records how many parts each machine is capable of running. When choosing which machine to assign a part to, the system always prefers the machine with the fewest compatible parts. A machine that runs only two parts is rated higher than a machine that runs fifteen parts. This ensures that specialised machines are claimed by their dedicated parts before any general part can occupy them.

Figure 4 shows an example of how the machine ranking works.

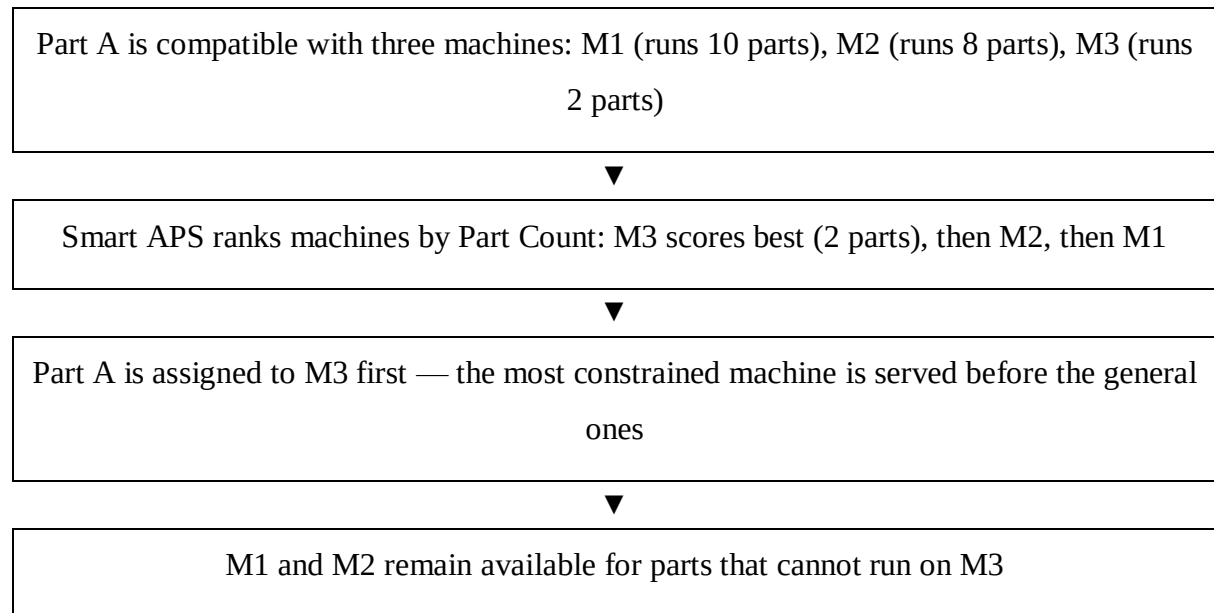


Figure 4: Specialised machine protection through Part Count ranking.

If It Still Fails

If even the specialised machine is fully occupied and a dedicated part cannot be planned, it appears in the VT_Not_Planned report with the reason clearly stated. The planner can then review the specialised machine's schedule and free up hours if necessary.

7. CONSTRAINT 6: USE THE MINIMUM NUMBER OF MOULDS

The Problem

Each part has a physical mould (tool) that must be fitted to a machine to produce it. Some parts have more than one mould available. Using too many moulds for a single part simultaneously ties up multiple machines unnecessarily, reducing the capacity available for other parts. The goal is always to produce the required quantity using as few machines as possible.

Our Solution

Smart APS always attempts to produce the full daily requirement for a part on a single machine first. Only if that machine does not have enough hours left in the shift does the system bring in a second machine. When a second machine is used, it only runs for the hours needed to cover the shortfall from the first machine, not the full requirement again. A third machine is only used if both the first and second together are still insufficient. Figure 5 shows this decision process.

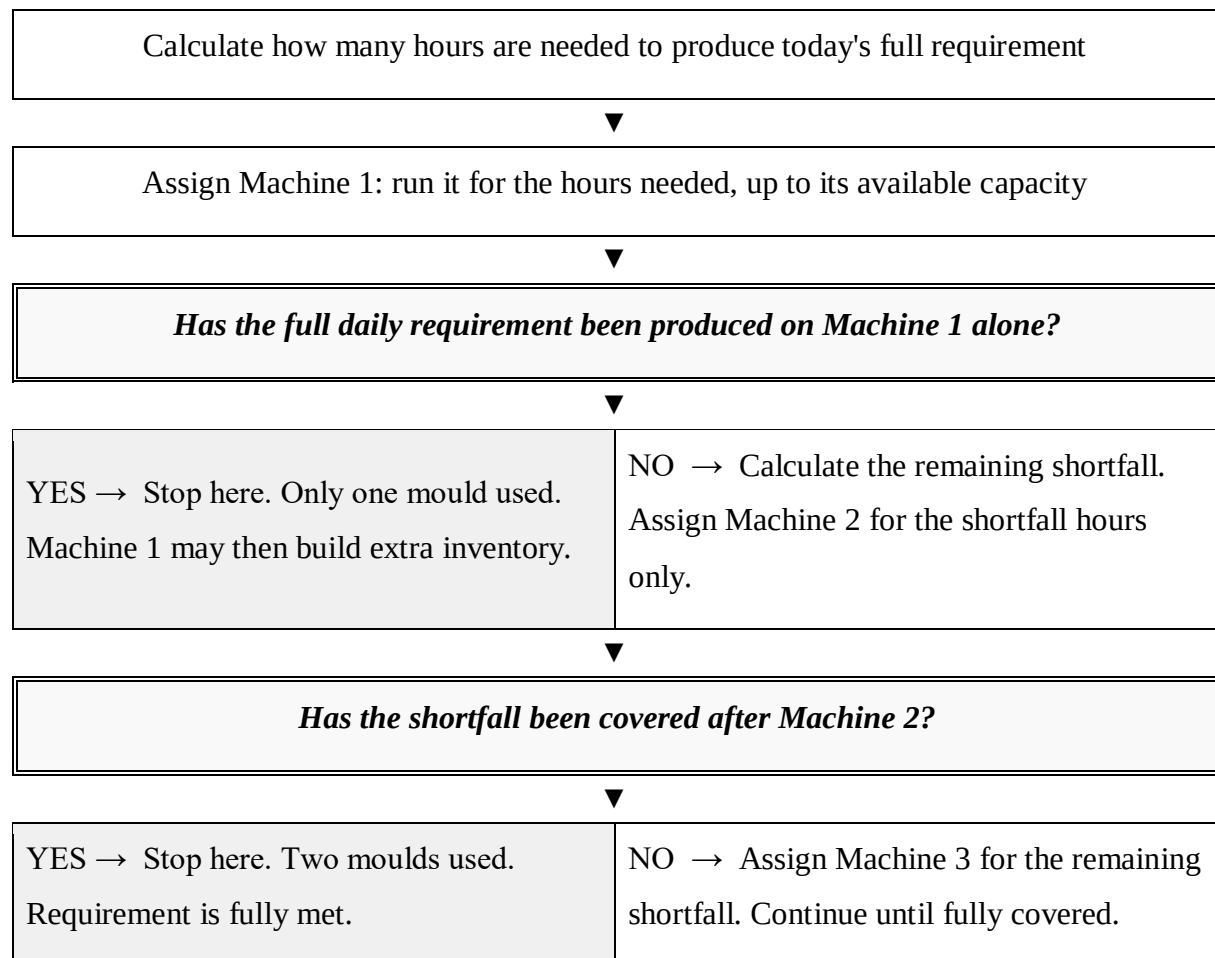


Figure 5: Minimum tools logic — each additional machine covers only the remaining shortfall.

If It Still Fails

If the full daily requirement cannot be met even with all available moulds deployed, the system reports the gap in the VT_Production_vs_Indent sheet, showing exactly how much was produced versus how much was required. The planner can then decide whether the shortfall can be covered from existing inventory.

8. CONSTRAINT 7: PLAN QUANTITIES BASED ON INVENTORY HEALTH

The Problem

Not every day is the same. On some days, all parts have several days of stock sitting in the warehouse and the situation is comfortable. On other days, several parts are at zero stock and any delay in production could stop the supply chain. If the planning system treats both situations identically, it will either over-produce when things are healthy or under-react when there is a crisis.

Our Solution

Every morning, before building the plan, Smart APS assesses the inventory health of all active parts and classifies the day into one of four situations, which are called scenarios. The planned quantity for each part is then set according to the scenario. Table 2 describes the four scenarios.

Scenario	Condition	What the System Does	Quantity Cap
S0 — Crisis	All parts have less than one day of stock	Produce just enough to survive today with a small safety margin. Machine time is carefully rationed across all parts.	1.5 times daily requirement
S1 — Mixed	Some parts are critical; others have stock	Critical parts are produced first and filled to a three-day buffer. Non-critical parts are deferred if necessary.	3 times daily requirement
S2 — Recovery	All parts exist but are below the three-day target	Produce enough to restore the three-day inventory buffer for all parts.	4 times daily requirement
S3 — Healthy	All parts have three or more days of stock	Cycle through all parts systematically, producing	5 times daily requirement

		approximately one day of supply for each in turn.	
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Table 2: The four production scenarios and their corresponding planning logic.

If It Still Fails

If despite the scenario-adjusted planning some parts still cannot be produced due to machine capacity constraints, they appear in the VT_Not_Planned report. The current scenario is printed clearly at the top of every output sheet so the planner always understands the overall factory situation at a glance.

9. CONSTRAINT 8: MOST URGENT PARTS MUST BE PLANNED FIRST

The Problem

If parts are assigned to machines in a random or arbitrary order, a critically urgent part with zero inventory might end up being scheduled after a less important part. By the time the urgent part reaches the front of the queue, all suitable machines may already be fully occupied. The most critical part misses the day entirely, which is the worst possible outcome.

Our Solution

Before any scheduling begins, Smart APS calculates a priority score between zero and one hundred for every active part. Parts are then assigned to machines in order of their score, from highest to lowest. This guarantees that the most urgent part is always the first to receive a machine. The score is based on three factors, as shown in Table 3.

Factor	Weight in Score	How It Is Calculated
Urgency: how far below the three-day inventory target is this part?	55 percent	A part with zero stock scores the full 55 points. A part with a full three-day buffer scores zero on this factor.
Category: is it a Runner, Repeater, or Stranger?	25 percent	A Runner (daily high-frequency part) scores 25 points. A Repeater scores 15. A Stranger scores 5.
Relative demand: how much of this part is needed compared to all others?	20 percent	The part with the highest daily demand scores the full 20 points. Others are scored proportionally.

Table 3: Priority score components and their weightings.

A part with zero inventory that is also a Runner will score close to 100 out of 100, and will always be among the first parts assigned to a machine, regardless of any other consideration.

If It Still Fails

Even with the highest possible priority score, a part may not be planned if all compatible machines are genuinely full. In this case, the part appears at the top of the VT_Not_Planned report with its score clearly visible, so the planner can immediately see what is most urgent and take action.

10. CONSTRAINT 9: NO MACHINE SHOULD SIT IDLE

The Problem

After all the urgent parts have been planned, some machines will have hours remaining in their shift. If nothing is done with this remaining capacity, those hours are simply wasted. Across 30 machines, even a small amount of idle time per machine per day adds up to a significant loss of production capacity over time.

Our Solution

After the main planning pass is complete, Smart APS runs a second pass called the Utilisation Enforcer. It checks every machine for remaining free hours and attempts to fill them in two steps, as shown in Figure 6.

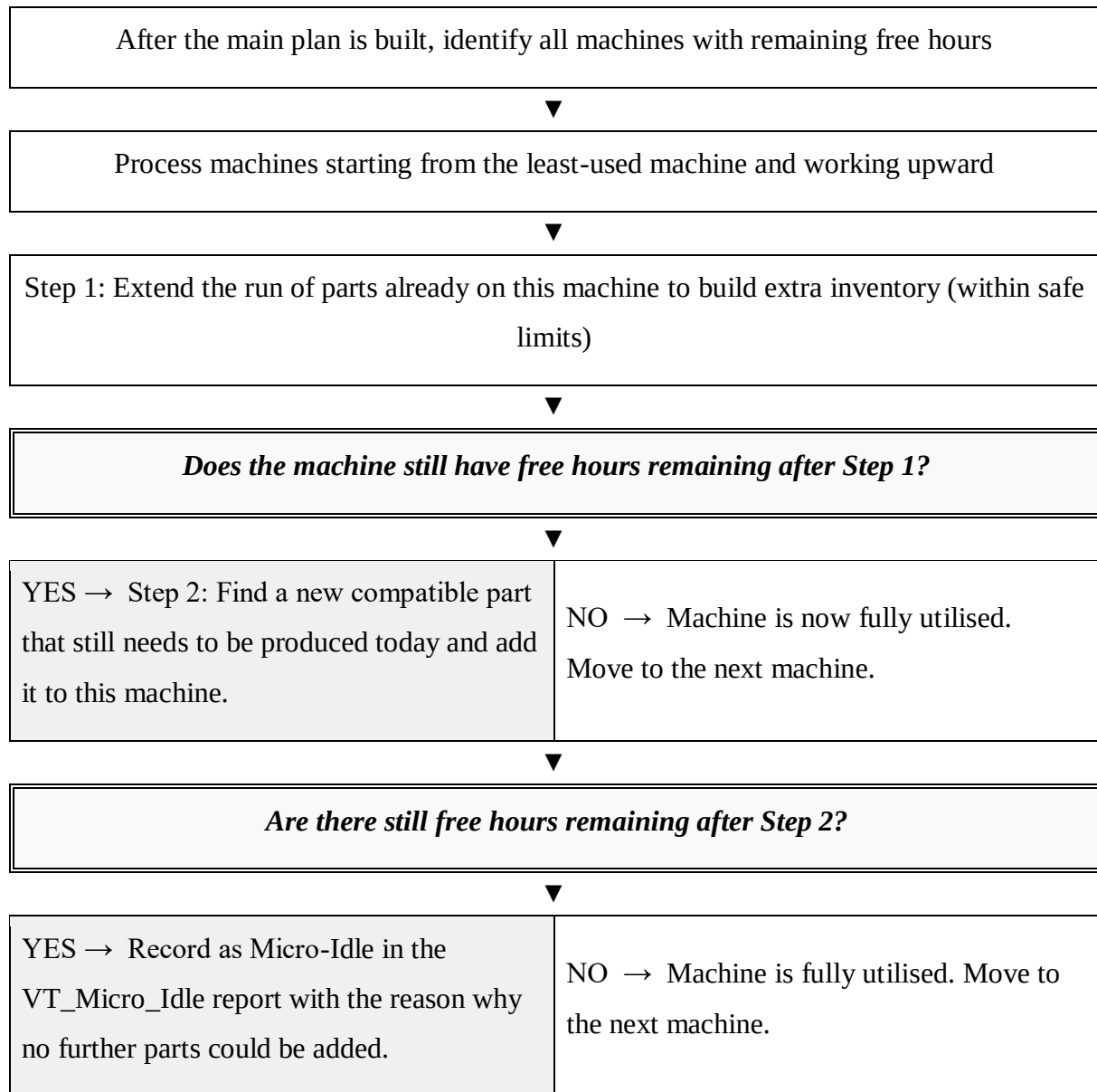


Figure 6: The Utilisation Enforcer — two-step process for filling idle machine time.

If It Still Fails

If a machine still has idle time after both steps, it means all compatible parts have either already been planned or have reached their maximum allowable production quantity for the day. This residual idle time is logged in the VT_Micro_Idle sheet with a full explanation, giving management complete visibility of why the machine could not be filled.

11. SUMMARY OF ALL CONSTRAINTS

Table 4 below provides a concise reference summary of all nine constraints, their solutions, and their fallback actions.

No.	Constraint	Solution	Fallback
1	Machine cannot exceed 22 hours per shift	Hard ceiling: available hours tracked at all times.	Part logged in VT_Not_Planned with free-hours detail.
2	Minimum four-hour run block per part	Part only assigned if four or more effective hours remain after changeover.	Next compatible machine is tried. If none, part goes to VT_Not_Planned.
3	One tool changer for all 30 machines	CSP algorithm sequences all changeovers so no two overlap. Machine waits are filled with extended production.	Ten-second timeout triggers fast greedy method. No-overlap rule is still maintained.
4	Continuing the same part avoids a changeover	Machine state saved daily. Same-part continuation strongly preferred in machine ranking.	If unavailable, changeover is charged and sequenced by the tool-changer scheduler.
5	Specialised machines must not be taken by general parts	Part Count ranking: machine with fewest compatible parts is always preferred first.	If specialised machine is full, appears in VT_Not_Planned for planner action.
6	Use the minimum number of moulds	Try one machine first. Add a second only for the remaining shortfall. Repeat if necessary.	Gap between required and produced shown in VT_Production_vs_Indent report.

7	Quantities depend on inventory health	Morning scenario classification (S0 to S3) sets the quantity logic for all parts.	Scenario and reason printed on every output sheet for planner awareness.
8	Most urgent parts must be planned first	Priority score (0 to 100) computed per part. Parts assigned in score order.	Highest-priority unplanned parts shown at the top of VT_Not_Planned.
9	No machine should sit idle	Utilisation Enforcer extends existing runs, then adds new parts to fill remaining hours.	Residual idle time logged in VT_Micro_Idle with the reason.

Table 4: Summary of all nine constraints, solutions, and fallback actions.

12. APPROVAL AND SIGN-OFF

This document has been prepared to ensure that all stakeholders, irrespective of their technical background, have a complete understanding of the rules governing the Smart APS system before it is deployed for daily production use.

By signing below, the authorised personnel confirm that they have read and understood the constraints described in this document and approve the Smart APS V8 system for use as the primary production planning tool for the VT Machine Line in the Injection Moulding Division.

Prepared By (Intern): Tushar:

Name: _____

Date: _____

Guided By: Mr. Amrit Lal:

Name: _____

Date: _____

Approved By: Plant Manager / VP Operations:

Name: _____

Date: _____

Document Version History

Version	Date	Prepared By	Summary of Changes
V1.0	March 2026	Tushar, guided by Mr. Amrit Lal	Initial version for V8 approval